Is the compass qubit a noiseless subsystem?

What `ns_compass.py` computes, how, and how to read the output

Notes for the two-photon Dicke thesis project

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In short

`ns_compass.py` takes the dissipative two-photon Dicke model of the thesis (Eqs. 4.9–4.10, two-photon loss only) and asks, without any mean-field input: *do the eight slow Liouvillian modes of the superradiant (SR) phase have the structure of two qubits stored as noiseless subsystems (NS), and how are those two qubits related?* For every system size N it

1. cuts the Liouvillian into the eight blocks of the nested symmetry $\mathbb{Z}_4^W \supset \mathbb{Z}_2^S$ (same labels as `sweep.py`);
2. computes in each block the slowest right *and* left eigenmodes;
3. removes the vacuum (normal-phase) part of the bistable steady state;
4. superimposes the weak-$\mathbb{Z}_4$ partner blocks to separate the two compass qubits A and B;
5. applies the NS test of Lieu *et al.* [PRL 125, 240405 (2020), SM Sec. 2 and 4] to each qubit;
6. quantifies how the two qubits relate.

What the test runs on this machine showed ($N \leq 8$, three parameter sets, Sec. 8):

- In every block there is exactly *one* slow superradiant mode, separated by a gap from all others. The slow manifold is 8-dimensional: a direct sum of two qubits, not a 4-level system (16 modes) and not a single qubit.
- Each qubit passes the NS test: the four steady operators of qubit A (and of B) have the form $|\mu\rangle\langle\nu| \otimes M$. At the thesis parameters ($\lambda = 0.6$, $\kappa_2 = 0.05$) the trace distances of the test decrease with N, to 2.5×10^{-5} (populations) and 1.9×10^{-4} (coherence) at $N = 5$. M is mixed (purity 0.78), so it is a noiseless subsystem, not a decoherence-free subspace. The conserved quantity of the coherence is the identity on the dominant auxiliary states to 7×10^{-4}.
- The two qubits are images of each other under Π_4, live on orthogonal supports (spins along $+x$ versus $-x$, photon lobes rotated by 90°), share no coherence, and the environment destroys any coherence between them 100–600 times faster than coherence inside one qubit (the factor grows with N). The steady state is a classical mixture of two qubits, $\rho = p\,\rho_A \otimes M_A \oplus (1-p)\,\rho_B \otimes M_B$.
- **Finite-size caveat.** At accessible N the eight modes are only *metastable*: at the thesis parameters their decay rates stay at $\sim 10^{-4}\omega_c$ from $N = 3$ to $N = 5$ instead of falling exponentially with N. The data point to collective spin flips: superradiance needs a spin polarisation $|j_x| > \omega_c/2\lambda$, and for $\lambda = 0.6$ one single flip already pushes the spin below that threshold for every $N \leq 11$. Where more flips are needed ($\lambda = 1$, $\kappa_2 = 0.5$, where k^* rises from 1 to 3 between $N = 2$ and 8) the rates fall steadily with N, by a factor 4–6. The NS deviations track the rates (Sec. 7).
- **Side result for `sweep.py`.** Its cutoff $M = 120$ is converged at $N = 6$ only up to $\lambda \approx 0.54$; near $\lambda = 0.57$ the computed gap is not converged (Sec. 2, Fock cutoff).

1 The structure being tested

Model and blocks. $H = \omega_c a^\dagger a + \frac{\omega_a}{2} J_z + \frac{\lambda}{N} J_x (a^{\dagger 2} + a^2)$ with $J_{x,z}$ twice the standard spin-$N/2$ operators, and one jump operator $L_2 = \sqrt{\kappa_2/N}\, a^2$. The photon parity $\Pi_2 = e^{i\pi a^\dagger a}$ commutes with H and L_2 (strong $\mathbb{Z}_2$); $\Pi_4 = e^{i\pi(a^\dagger a/2 + J_z^{\mathrm{std}})}$ commutes with H but anticommutes with L_2 (weak $\mathbb{Z}_4$). A matrix element $|h\rangle\langle h'|$ belongs to block E^w_{lr} with l, r the photon parities of h, h' and $w = \pi_4(h)\pi_4(h')^* \in \{\pm 1, \pm i\}$; parity fixes w up to sign, which gives the eight blocks $E^{\pm 1}_{++}, E^{\pm 1}_{--}, E^{\pm i}_{+-}, E^{\pm i}_{-+}$. The script checks that the Liouvillian has no matrix element between blocks.

Expected steady states (thesis Sec. 4.5.1). In the SR phase there are two *pairs* of photon lobes. Pair A: lobes at $\pm\alpha$ with the collective spin along $+x$; pair B: lobes at $\pm i\alpha$ with the spin along $-x$ (Π_4 rotates the field by 90° and flips J_x). Inside each pair the photon parity defines a cat qubit, $|e_A\rangle \propto |\alpha\rangle + |-\alpha\rangle$, $|o_A\rangle \propto |\alpha\rangle - |-\alpha\rangle$ (thesis $|q_1\rangle, |q_3\rangle$), and $|e_B\rangle = \Pi_4|e_A\rangle$, $|o_B\rangle = \Pi_4|o_A\rangle$ ($|q_2\rangle, |q_4\rangle$). Qubit A is a noiseless subsystem if its four steady operators are

$$R^A_{\mu\nu} = W_\mu M W_\nu^\dagger, \qquad \mu, \nu \in \{e, o\}, \tag{1}$$

with isometries W_e (into the even sector) and W_o (into the odd sector) from an auxiliary space, and one density matrix M shared by all four. In words: the steady state is $\rho_L \otimes M$ with an *arbitrary* logical state ρ_L and a *fixed* auxiliary state M (photon and spin fluctuations around the lobes). If M is pure this is a decoherence-free subspace (DFS), if mixed an NS. Lieu *et al.* found an NS for the two-photon-driven cavity away from its special point; the question here is whether the same holds for each compass qubit.

What the eight blocks contain. No single block contains qubit A alone. With $R^B = \Pi_4 R^A \Pi_4^\dagger$:

block	slow mode (SR phase)	meaning
E^{+1}_{++}	$S_e = \frac{1}{2}(R^A_{ee} + R^B_{ee})$	even-parity steady state (exact zero)
E^{-1}_{++}	$T_e \propto R^A_{ee} - R^B_{ee}$	population imbalance A−B
$E^{\pm 1}_{--}$	same with o	odd sector
E^{+i}_{+-}	$C_+ \propto R^A_{eo} - iR^B_{eo}$	logical coherences
E^{-i}_{+-}	$C_- \propto R^A_{eo} + iR^B_{eo}$	
$E^{\mp i}_{-+}$	$C_\pm^\dagger$	Hermitian conjugates (not recomputed)

This is why the qubits must be isolated by superimposing blocks (Sec. 4). Note also what is *absent*: a coherence between the qubits, e.g. $|e_A\rangle\langle e_B|$, would add a *second* slow mode to $E^{\pm 1}_{++}$. A 4-level NS would have 16 slow modes, two per block; the direct sum of two qubits has 8, one per block. Counting slow modes per block therefore distinguishes the two cases.

2 Numerical method

Liouvillian. Built with QuTiP exactly as in `sweep.py` (`qt.liouvillian(H,[L2])`, column-stacked vectorisation, spin space $j = N/2$), then converted to a scipy sparse matrix and cut into blocks.

Slow modes. In every block the `N_MODES`= 6 eigenvalues closest to 0 are found by shift-invert Arnoldi (ARPACK) around $\sigma = -10^{-7} + 10^{-9}i$. One sparse LU factorisation of $(L_{\mathrm{block}} - \sigma)$ gives both the right eigenvectors ($Lr = \lambda r$, the steady-state candidates) and the left eigenvectors ($L^\dagger \ell = \lambda^* \ell$, the conserved quantities), through `lu.solve(x)` and `lu.solve(x, trans='H')`. The E_{-+} blocks follow from $\mathcal{L}(X^\dagger) = \mathcal{L}(X)^\dagger$ and are not recomputed (set `COMPUTE_CONJUGATE_BLOCKS=True` to check this numerically).

Ordering of the LU. SuperLU's default ordering (COLAMD) produces catastrophic fill-in for this problem: at $N = 6$ a single block did not fit in 7 GB. Every vectorised index is a point (n, s, n', s') (photon number and spin index of ket and bra) of a 4D lattice, and $\mathcal{L}$ only couples neighbouring points (n changes by 2, s by 1). The script orders the unknowns by geometric nested dissection (recursive bisection, separators last) and calls SuperLU with that fixed ordering. At $N = 4$ this halves the LU; at $N = 6$ the factorisation needs about 1 GB instead of more than 7 GB, which is what makes $N = 7$–8 feasible on the cluster.

Fock cutoff. The SR lobe sits at $n_{\mathrm{MF}} = N\sqrt{4\lambda^2 - \omega_c^2}/\kappa_2$ (thesis Eq. 4.51 with $|\langle J_x\rangle| \to 1$; at the thesis parameters the numerical mean photon number of the lobe is 2–6 % lower) and is about 1.5 times as wide as a coherent state. The default $M = n_{\mathrm{MF}} + 10\sqrt{n_{\mathrm{MF}}} + 6$ leaves a relative weight $\lesssim 10^{-10}$ of the slow modes in the top four Fock states, which the script reports (`tail weight`, with a warning above 10^{-7}). **This check matters:** an insufficient cutoff makes the symmetry-breaking modes decay artificially. At $N = 4$, $\lambda = 0.6$, $M = 80$ gave an E_{++}^{-1} eigenvalue of -5.5×10^{-3}; the converged value ($M \geq 110$) is -8.5×10^{-5}.

This also concerns the gap-closing curves of `sweep.py` ($N = 6$, $M = 120$, sector E_{--}^{-1}). For $N = 6$ the rule asks for $M = 126$ at $\lambda = 0.54$ and $M = 154$ at $\lambda = 0.57$. A direct check of that sector with $M = 120$ gives a tail weight of 1.7×10^{-7} at $\lambda = 0.54$ (eigenvalue -7.41×10^{-3}, against -7.62×10^{-3} with $M = 100$: converged to a few per cent), but 7.5×10^{-5} at $\lambda = 0.57$, where the eigenvalue moves from -4.3×10^{-3} ($M = 100$) to -2.5×10^{-4} ($M = 120$) and is clearly not converged. The $N = 6$ curve is therefore reliable up to $\lambda \approx 0.54$; closer to 0.57 the computed gap is likely too large.

Parallelisation and memory. The six independent blocks are solved in parallel worker processes (one LU each). The number running at once is limited by `--workers` and by `--mem-budget`, using the measured peak memory of a worker (≈ 4 GB at block dimension 8×10^4, growing like $\dim^{1.65}$). If a worker is killed anyway (out of memory), the script stops with an explicit message instead of hanging; the results of the smaller N are already saved, so the job can be restarted from the N that failed with fewer `--workers`.

3 Caveat 1: isolating the superradiant phase from the vacuum

The problem. The SR and normal phases coexist: the exact steady state in each parity sector is a mixture of SR lobes at $n \sim n_{\mathrm{MF}}$ and a vacuum part at small n (Fig. 4). The vacuum part is not an NS: its even and odd components ($|0\rangle\langle 0|$-like and $|1\rangle\langle 1|$-like) differ, so a Lieu test applied to the raw steady states would fail for reasons that have nothing to do with the SR phase. Keeping the first few Fock states indeed makes the NS distances 2–5 times worse at $N = 2, 3$ (Table 1 below).

What the script does.

1. It locates the SR lobe from the slow E_{++}^{-1} mode. That mode is the A−B population imbalance and exists only in the SR phase, so its photon-number profile marks the SR peak n_{peak} without any vacuum contamination.
2. It finds the valley of the exact steady-state distribution $P(n)$ between 0 and n_{peak}; its position is the cut n_{cut}.
3. Every selected mode is projected onto the SR basin,

$$r \to P_{\mathrm{SR}}\, r\, P_{\mathrm{SR}}, \qquad P_{\mathrm{SR}} = \sum_{n \geq n_{\mathrm{cut}}} |n\rangle\langle n| \otimes \mathbb{1}_{\mathrm{spin}}.$$

4. Mode selection. In E_{++}^{+1} and E_{--}^{+1} it takes the mode that carries the most SR population, $|\operatorname{Tr}(P_{\mathrm{SR}} r)|/\|r\|$. This is the exact steady state when the SR phase dominates it, otherwise

the vacuum↔SR switching mode; after the projection both give the SR state, while vacuum-internal and SR-internal relaxation modes carry no SR population and are never picked. In all other blocks it takes the slowest mode that lives mostly ($> 50\%$) in the SR basin.

Why this is justified, and the checks. The two phases are separated in photon number by a distance $\propto N$; at the thesis parameters $P(n)$ in the valley is 3×10^{-3} ($N = 3$) to 9×10^{-4} ($N = 5$) of its value at the peak. The script reports the valley depth, the steady-state weight below the cut, and repeats the whole analysis with the cut moved by ± 4 photons (Table 1). For $N \geq 4$ this changes the NS distances by less than a factor 2 in either direction. At $N = 2, 3$, where the lower value puts the cut at $n = 0$ and so keeps the vacuum, they become 2–5 times worse: the vacuum does contaminate the test, and the cut removes it.

Table 1: Vacuum-cut scan, thesis parameters, qubit A: NS distances with the cut at $n_{\rm cut} - 4$, $n_{\rm cut}$, $n_{\rm cut} + 4$ (cuts below 0 are set to 0).

N	cuts	$D_t(z_{ee}, z_{oo})$	$D_t(z_{ee}, z_{eo})$
2	0 / 2 / 6	4.7 / 1.4 / 1.8 ($\times 10^{-3}$)	9.1 / 2.8 / 1.7 ($\times 10^{-3}$)
3	0 / 2 / 6	4.6 / 1.0 / 1.6 ($\times 10^{-4}$)	15.0 / 7.4 / 5.4 ($\times 10^{-4}$)
4	6 / 10 / 14	7.0 / 4.9 / 7.9 ($\times 10^{-5}$)	4.7 / 2.5 / 1.9 ($\times 10^{-4}$)
5	12 / 16 / 20	2.9 / 2.5 / 3.4 ($\times 10^{-5}$)	2.8 / 1.9 / 2.0 ($\times 10^{-4}$)

Control: the vacuum is not what limits the slow modes. To test whether vacuum↔SR switching causes the finite lifetime of the eight modes, the model was also solved in a Fock space restricted to $n \geq n_{\rm cut}$ (the vacuum basin removed from the dynamics altogether). At $N = 3$, $\lambda = 0.7$ the E_{++}^{-1} rate changed only from 5.7×10^{-5} to 5.4×10^{-5}. So the vacuum has to be projected out to see the NS structure, but it is not the dominant leak at these sizes (Sec. 7). The restricted model is a control only; the production analysis always uses the exact model.

Alternatives considered. Preparing SR states and time-evolving them for an intermediate time (Lieu's quench protocol) needs full density-matrix propagation with a stiff generator and is far more expensive than one LU per block. Extreme metastable states built from left eigenvectors (Macieszczak *et al.*, PRL 116, 240404) work only with a clean separation of time scales, which the vacuum basin, full of slow spin and photon modes, does not provide here.

4 Caveat 2: separating the two qubits

Supports. Because R^A and R^B live on orthogonal subspaces, the Hermitian operator T_μ is positive on qubit A's support and negative on qubit B's. The script diagonalises T_μ and builds the projectors P_μ^A (P_μ^B) onto its positive (negative) eigenvectors above 10^{-10} of the largest eigenvalue. Labels are fixed physically: A is the pair whose spins point along $+x$, $\langle J_x \rangle > 0$, in both parity sectors.

Superposition coefficients. Qubit A's operators are superpositions of block modes, with coefficients fixed by demanding *no weight on qubit B's support*:

$$\rho_{\mu\mu}^A \propto S_\mu + c_\mu T_\mu, \qquad c_\mu = -\frac{\mathrm{Tr}(P_\mu^B S_\mu)}{\mathrm{Tr}(P_\mu^B T_\mu)}, \tag{2}$$

$$\rho_{eo}^A \propto C_+ + r\, C_-, \qquad r = -\frac{\langle B_-, B_+ \rangle}{\langle B_-, B_- \rangle}, \quad B_\pm = P_e^B C_\pm P_o^B, \tag{3}$$

and the same with A and B exchanged. Diagnostics printed: the residual of the B-part cancellation, the weight of S_μ outside both supports, the coherence between the supports left in S_μ, and the *shape*

mismatch D_t between T_μ and S_μ restricted to one qubit (is the E^{-1} mode really the imbalance of the *same* two states?).

5 The noiseless-subsystem test

For qubit A (and B) the script follows Lieu *et al.*, SM Sec. 2: diagonalise $\rho^A_{ee} = U_e z_{ee} U_e^\dagger$ and $\rho^A_{oo} = U_o z_{oo} U_o^\dagger$ (eigenvalues sorted) and transform the coherence, $z_{eo} = U_e^\dagger \rho^A_{eo} U_o$. The column phases of U_o are a gauge freedom; they are fixed so that $\mathrm{diag}(z_{eo})$ is real and positive, and z_{eo} is normalised to unit trace. For Eq. (1)

$$z_{ee} = z_{oo} = z_{eo} = \mathrm{diag}(m_1, m_2, \dots) \quad \text{(spectrum of } M\text{)}.$$

Reported: $D_t(z_{ee}, z_{oo})$, $D_t(z_{ee}, z_{eo})$ (both $\frac{1}{2}\,\mathrm{Tr}\,|\cdot|$ over the resolved part, eigenvalues $> 10^{-12}$), the largest off-diagonal element of z_{eo}, and the purity $\mathrm{Tr}\, z_{ee}^2$ (NS if < 1, DFS if $= 1$). The coherence is estimated twice: from the superposition above (main) and from E^{-i}_{+-} alone (check); at the thesis parameters the two values of $D_t(z_{ee}, z_{eo})$ agree to within 30 % at $N = 2$ and 4 % at $N = 5$.

Conserved quantities (SM Sec. 4). An NS stores information only if the conserved quantity of the coherence mode is the identity on the code, $\ell_{eo} = W_e W_o^\dagger$. The left eigenvector of the coherence mode is transformed with the same U_e, U_o and phases, normalised by $\mathrm{Tr}(q^\dagger z_{eo}) = 1$, and the script reports $\max |q_{kl} - \delta_{kl}|$ over the auxiliary states with $z_k > 10^{-3}$. States with smaller weight are poorly resolved at finite N (Lieu make the same remark); Fig. 2(c) shows the full matrix.

6 How the two noiseless subsystems relate

1. **Direct sum, no shared coherence.** One slow SR mode per block, with the next SR mode a gap away (Fig. 1b). Consistently, the coherence between the A and B supports in the steady state, and the A–B cross terms inside the coherence modes, are small and fall quickly with N (Fig. 3a). The script also inspects every computed mode of every block: a mode counts as A–B coherence if most of its weight on the dominant auxiliary states of the two qubits (weight $> 10^{-3}$, restricted to each qubit's own support) sits in the A–B off-diagonal part. It reports the slowest such mode, or a lower bound on its rate if there is none.

2. **Why.** The environment sees a^2. On the lobes $a^2 \approx \alpha^2$ for pair A but $-\alpha^2$ for pair B, while even and odd cats of the same pair have the same a^2. For $X = |u\rangle\langle v|$ the instantaneous decay rate is
$$\Gamma(u, v) = \tfrac{1}{2}\big(\langle L_2^\dagger L_2\rangle_u + \langle L_2^\dagger L_2\rangle_v\big) - \mathrm{Re}\,\langle L_2\rangle_u \langle L_2\rangle_v^*,$$
which the script evaluates for the dominant states of each qubit. Coherence inside a qubit decays exactly like a population (the environment cannot tell e from o), coherence between the qubits $\sim 2(\kappa_2/N)|\alpha|^4$ faster, i.e. 95–630 times faster in the runs below, and the ratio grows with N (as $|\alpha|^4/N \propto N$).

3. **Same states, mapped by Π_4.** Π_4 maps qubit A onto qubit B with the same auxiliary state; the script verifies $\Pi_4 \rho^A \Pi_4^\dagger = \rho^B$ to machine precision. $\Pi_4^2 = \pm\Pi_2$ acts on each qubit as the logical Z. So the *strong* $\mathbb{Z}_2$ labels the logical basis of *both* qubits at once (this is the sense in which both are built from the same parity sectors), while the *weak* $\mathbb{Z}_4$ swaps the qubits.

4. **The which-qubit bit.** The left eigenvector of the E^{-1}_{++} mode is the conserved quantity that remembers which qubit is occupied: $+1$ on A, -1 on B. This is the classical bit of the broken weak symmetry; the qubits themselves come from the broken strong symmetry.

Together: the steady states of the broken phase are $\rho = p\,\rho_A \otimes M \oplus (1-p)\,\Pi_4(\rho_B \otimes M)\Pi_4^\dagger$ with two arbitrary qubit states $\rho_{A,B}$ and a probability p: $3 + 3 + 1 = 7$ real parameters, i.e. the eight slow modes minus the trace. This is the "statistical mixture of two qubits rotated by 90°" anticipated in thesis Sec. 4.2.3, now obtained from the exact Liouvillian.

7 Finite-size behaviour: what limits the NS at accessible N

Observation. At the thesis parameters the decay rates of the eight modes stay at $\sim 10^{-4}\omega_c$ from $N = 3$ to $N = 5$, and $D_t(z_{ee}, z_{eo})$ follows them (Tables 2 and 3). A slowly decaying coherence mode cannot be exactly of the form (1); the deviation is of the order of its rate divided by the SR gap.

Mechanism. The photon mode feels the two-photon drive $\lambda j_x(a^2 + a^{\dagger 2})$, where the spin polarisation is $j_x = \langle J_x^{\mathrm{std}}\rangle/(N/2) \in [-1, 1]$. Superradiance is sustained by the parametric instability of this drive, which requires $2\lambda|j_x| > \omega_c$ (thesis Eq. 4.51 at $\langle n\rangle \to 0$). Starting from the fully polarised spin, the collapse therefore needs

$$k^* = \left\lfloor \tfrac{N}{2}\left(1 - \tfrac{\omega_c}{2\lambda}\right)\right\rfloor + 1$$

collective spin flips. For $\lambda = 0.6$, $k^* = 1$ for all $N \le 11$: one spin flip from $m_x = N/2$ already switches superradiance off, and the system returns to A or B at random. The script measures this directly: the diagonal of the which-qubit conserved quantity in the basis $|n\rangle \otimes |J_x = m_x\rangle$ at the SR peak (Fig. 5) is 1 at full polarisation and drops to 0.33 or less after a single flip whenever $k^* = 1$ (all runs at $\lambda = 0.6$ and 0.7, and $N = 2, 3$ at $\lambda = 1$; at $N = 2$ it is exactly 0, since one flip leaves $j_x = 0$). Plotted against $j_x = m_x/(N/2)$ the curves for different N nearly coincide (Fig. 5). What changes with N is how many flips are needed to reach a given j_x; this is what should suppress the leak at large N, where losing the which-qubit information requires a macroscopic number of flips.

Evidence that this is the leak. (i) Removing the vacuum from the dynamics does not change the rates (Sec. 3). (ii) For $\lambda = 1$, $\kappa_2 = 0.5$, where k^* grows from 1 to 3 between $N = 2$ and $N = 8$, the rates fall monotonically by a factor 4–6 and the memory after one flip rises from 0 to 0.62 (Table 4). (iii) For $\lambda = 0.7$ the rates are lower than for $\lambda = 0.6$ but again flat while $k^* = 1$.

Consequence for the thesis. The NS structure itself is clearly present; the remaining deviations measure how metastable the broken phase is at finite N. To *see* the thermodynamic limit numerically, choose parameters where k^* increases inside the feasible range of N (larger λ), and keep κ_2 large enough that the photon number, hence M, stays manageable. A thesis-parameter run on the cluster ($N \le 8$, $k^* = 1$ throughout) should therefore show the same plateau; that would not indicate the absence of an NS.

8 Results of the test runs

All runs below were done with the delivered script on a 2-core, 7 GB machine, so they stop at small N. The cluster runs of Sec. 9 extend them.

Thesis parameters, $\lambda = 0.6$, $\kappa_2 = 0.05$, $\omega_a = \omega_c = 1$. Tables 2–3 and Figs. 1–4. The Fock tail weight was $< 3 \times 10^{-11}$ and the block-structure check gave 0 in all runs. A and B give identical numbers, as required by the Π_4 symmetry (the script prints both).

Table 2: Spectrum and vacuum isolation, thesis parameters. $|\lambda|$: decay rate of the selected slow mode in E^{-1}_{++} (A/B imbalance) and E^{i}_{+-} (coherence); ratio: second-slowest over slowest eigenvalue, minimum over the four symmetry-breaking blocks; last column: vacuum↔SR switching mode in E^{+1}_{++}.

N	M	k^*	n_{cut}	vac. weight	valley	$\lvert\lambda\rvert\, E^{-1}_{++}$	$\lvert\lambda\rvert\, E^{i}_{+-}$	ratio	switching
2	86	1	2	4.7×10^{-3}	2.8×10^{-2}	1.8×10^{-3}	7.5×10^{-4}	358	1.1×10^{-1}
3	110	1	2	5.1×10^{-4}	3.4×10^{-3}	1.8×10^{-4}	1.7×10^{-4}	2160	4.0×10^{-2}
4	132	1	10	9.5×10^{-4}	1.6×10^{-3}	8.5×10^{-5}	1.4×10^{-4}	1930	1.3×10^{-2}
5	154	1	16	1.6×10^{-3}	8.8×10^{-4}	9.1×10^{-5}	1.1×10^{-4}	1412	4.3×10^{-3}

Table 3: NS test and relation between the qubits, thesis parameters (qubit A; B is identical). q-err: conserved quantity vs. identity on auxiliary states with $z_k > 10^{-3}$; which-qubit: conserved quantity of E^{-1}_{++} vs. ± 1; A–B coh.: coherence between the supports left in S_e; Γ ratio: inter- over intra-qubit decay rate; $\ell(1)$: which-qubit memory after one spin flip.

N	$D_t(z_{ee}, z_{oo})$	$D_t(z_{ee}, z_{eo})$	purity	q-err	which-qubit	A–B coh.	Γ ratio	$\ell(1)$
2	1.4×10^{-3}	2.8×10^{-3}	0.755	1.0×10^{0}	2.0×10^{-1}	1.3×10^{-3}	95	0.00
3	1.0×10^{-4}	7.4×10^{-4}	0.770	8.3×10^{-3}	1.8×10^{-2}	1.5×10^{-4}	147	0.30
4	4.9×10^{-5}	2.5×10^{-4}	0.778	2.3×10^{-3}	2.3×10^{-3}	1.4×10^{-5}	199	0.33
5	2.5×10^{-5}	1.9×10^{-4}	0.782	6.5×10^{-4}	4.2×10^{-4}	1.7×10^{-6}	251	0.26

What the numbers say:

- **One slow mode per block.** In each of the four symmetry-breaking blocks the selected mode is 350–2000 times slower than the next one. In E^{+1}_{++}, E^{+1}_{--} the slow mode is the exact steady state and the next one the vacuum↔SR switching mode. At no N does any of the six computed modes per block carry coherence between the qubits, so a mode that does decays faster than the largest computed rate: $|\lambda| > 0.83, 0.63, 0.29, 0.21$ for $N = 2, 3, 4, 5$, i.e. at least 460 ($N = 2$) and about 2000 ($N = 3$–5) times faster than the slow modes.
- **NS structure.** $D_t(z_{ee}, z_{oo})$ falls by a factor 13 from $N = 2$ to 3 and by 2 per added spin after that, to 2.5×10^{-5} at $N = 5$; $D_t(z_{ee}, z_{eo})$ falls to 1.9×10^{-4}. The auxiliary state is mixed (purity 0.78): a noiseless subsystem, not a DFS. The conserved quantity of the coherence mode equals the identity on the four dominant auxiliary states to 7×10^{-4} at $N = 5$ (Fig. 2c shows where it fails: on states with weight $\lesssim 10^{-4}$). $N = 2$ is too small for the conserved quantities: there the q-error of 1.0 comes from the fifth auxiliary state ($z_5 = 2 \times 10^{-3}$), and the which-qubit observable still deviates from ± 1 by 0.2.
- **Two qubits, no shared coherence.** The A–B coherence left in the steady state drops by ~ 10 per spin (1.7×10^{-6} at $N = 5$); the which-qubit observable is ± 1 to 4×10^{-4}; inter-qubit coherence decays 95–250 times faster than intra-qubit coherence, and the ratio grows with N. $\langle J_x \rangle = \pm 0.999\, N/2$ on A/B, and $\langle a^2 \rangle$ flips sign between them.
- **Finite size.** The slow rates stay at $\approx 10^{-4}$, and $D_t(z_{ee}, z_{eo})$ flattens with them; $k^* = 1$ throughout (Sec. 7).

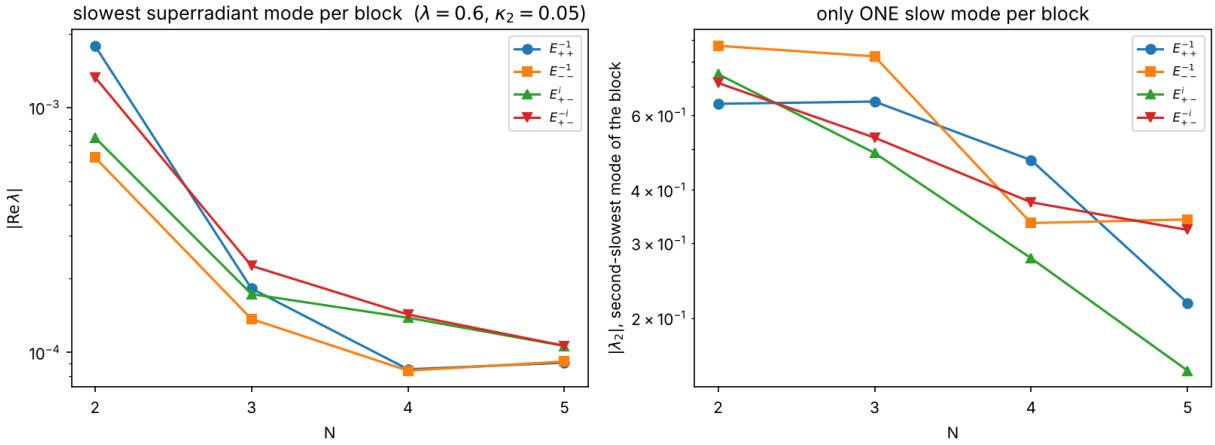


Figure 1: Spectrum, thesis parameters. Left: decay rate of the selected slow mode in the four symmetry-breaking blocks. Right: the second-slowest eigenvalue of the same block, 350–2000 times larger.

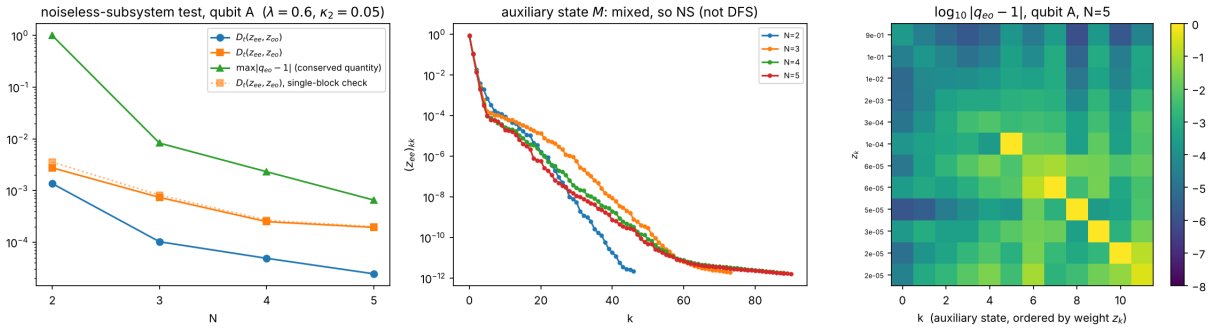


Figure 2: NS test, thesis parameters, qubit A. (a) Trace distances of Lieu *et al.* and the conserved-quantity error; the dotted line is the independent single-block estimate of the coherence. (b) Spectrum of the auxiliary state M (the floor at 10^{-11} is numerical). (c) $\log_{10}|q_{eo} - 1|$ for $N = 5$; rows are labelled with the weight z_k of the auxiliary state.

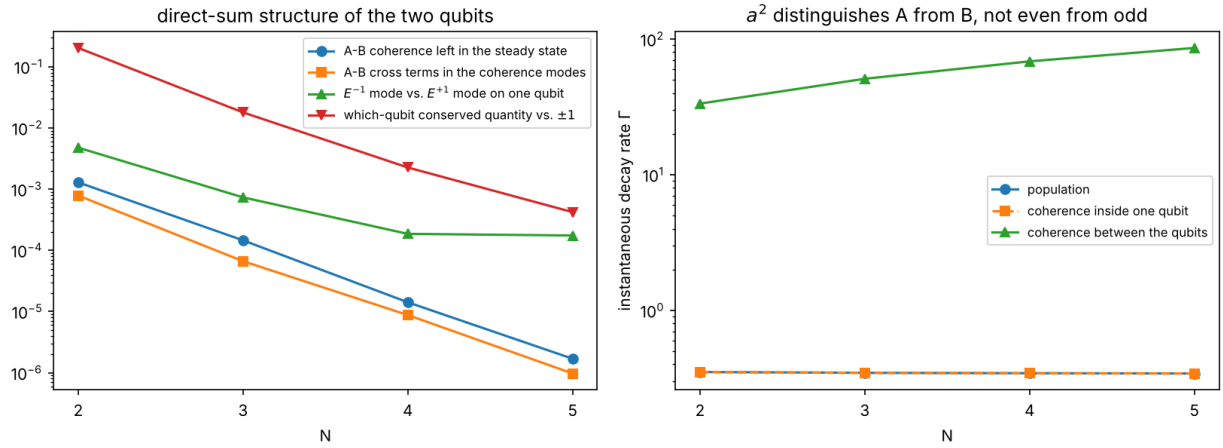


Figure 3: How the two qubits relate, thesis parameters. Left: everything that would signal coherence or mixing between the qubits vanishes with N (the E^{-1}/E^{+1} shape mismatch saturates with the slow rates). Right: inter-qubit coherence decays $\sim 10^2$ times faster than coherence inside one qubit, which decays exactly like a population.

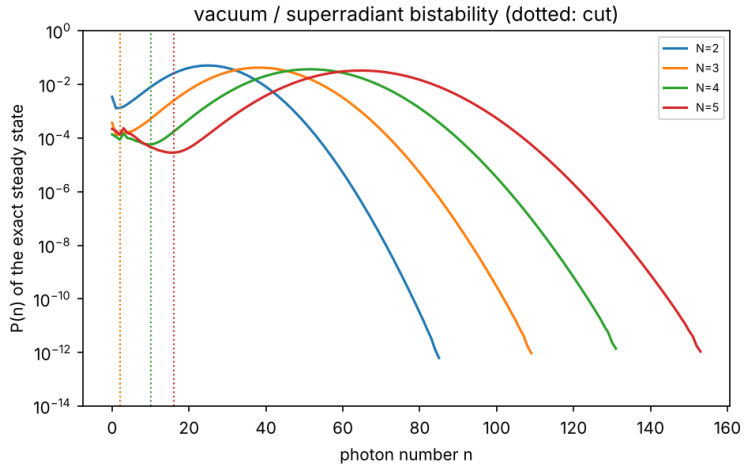


Figure 4: Photon distribution of the exact steady state (even and odd sector averaged) with the vacuum cut (dotted). The vacuum bump at $n \lesssim 10$ and the superradiant lobe are separated by a valley that deepens with N.

$\lambda = 0.7$, $\kappa_2 = 0.05$, $N = 2$–4. With 1.5 times as many photons per spin ($n_{\mathrm{MF}} \approx 20N$ instead of $13N$) the NS distances are 3–40 times smaller than at $\lambda = 0.6$ for the same N: $D_t(z_{ee}, z_{oo}) = 3.3\times10^{-5}$, 7.7×10^{-6}, 4.5×10^{-6} and $D_t(z_{ee}, z_{eo}) = 3.6\times10^{-4}$, 8.1×10^{-5}, 7.4×10^{-5} for $N = 2, 3, 4$; the A–B coherence in the steady state drops to 8×10^{-8} at $N = 4$. The slow rates again saturate ($|\lambda| = 5.6 \times 10^{-5}$ in E^{-1}_{++} and 8.3×10^{-5} in E^{i}_{+-} at both $N = 3$ and 4), and $D_t(z_{ee}, z_{eo})$ with them; $k^* = 1$ and $\ell(1) = 0.25$.

$\lambda = 1.0$, $\kappa_2 = 0.5$, $N = 2$–8. Here k^* grows from 1 to 3 (Table 4, Fig. 5). The photon number is small ($\approx 3.5N$), so vacuum and lobe overlap (valley ratio 0.16–0.4), and the slow modes are only 8–16 times slower than the next eigenvalue of their block. The NS distances are therefore at the per-cent level, the conserved-quantity tests (q-error, which-qubit observable) fail, and at some N a computed mode carries A–B coherence (decaying 14–40 times faster than the slow modes). These runs say little about the NS structure itself. Their point is the trend: the slow rates of all four symmetry-breaking blocks fall monotonically with N instead of saturating, by a factor 4–6 between $N = 2$ and 8, and the memory after one spin flip rises from 0 to 0.62.

Table 4: $\lambda = 1.0$, $\kappa_2 = 0.5$: the slow rates fall steadily as k^* grows, and the which-qubit memory after one spin flip, $\ell(1)$, rises. Ratio as in Table 2. The NS distances stay at the per-cent level (see text).

N	M	k^*	$\lvert\lambda\rvert\, E^{-1}_{++}$	$\lvert\lambda\rvert\, E^{i}_{+-}$	ratio	$D_t(z_{ee}, z_{oo})$	$D_t(z_{ee}, z_{eo})$	$\ell(1)$
2	40	1	4.9×10^{-2}	6.0×10^{-2}	16	2.8×10^{-2}	3.9×10^{-2}	0.00
3	50	1	4.3×10^{-2}	4.9×10^{-2}	13	4.1×10^{-2}	8.6×10^{-2}	0.19
4	58	2	3.9×10^{-2}	3.7×10^{-2}	10	3.0×10^{-2}	3.2×10^{-2}	0.21
5	66	2	2.7×10^{-2}	2.4×10^{-2}	8	4.9×10^{-2}	7.5×10^{-2}	0.40
6	74	2	1.7×10^{-2}	1.5×10^{-2}	8	3.6×10^{-2}	7.5×10^{-2}	0.52
7	80	2	1.3×10^{-2}	1.2×10^{-2}	8	1.9×10^{-2}	3.8×10^{-2}	0.58
8	88	3	1.1×10^{-2}	1.0×10^{-2}	8	1.2×10^{-2}	2.5×10^{-2}	0.62

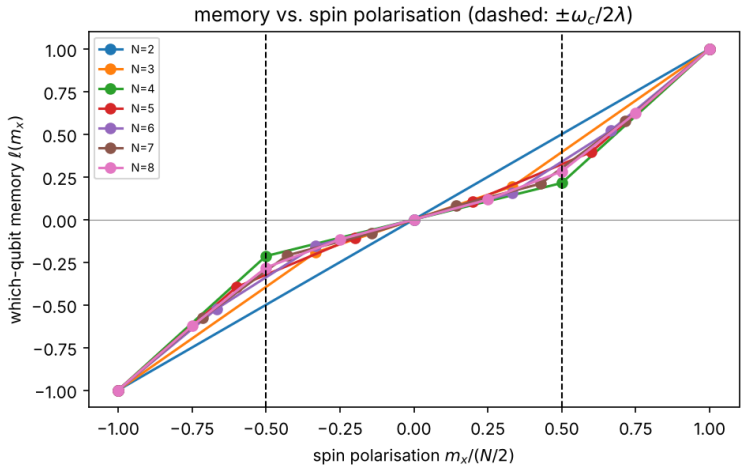


Figure 5: Which-qubit memory ℓ (diagonal of the conserved quantity of E^{-1}_{++} at the superradiant peak) versus the spin polarisation $m_x/(N/2)$, $\lambda = 1$, $\kappa_2 = 0.5$. $\ell = \pm1$: the system certainly returns to qubit A/B; $\ell = 0$: at random. For $N \geq 4$ the curves nearly collapse onto one function of $m_x/(N/2)$, which falls from 1 at full polarisation to 0.2–0.3 at the mean-field threshold $\omega_c/2\lambda$ (dashed) and to 0 at $m_x = 0$. Larger N needs more spin flips to get there.

9 Running it on the cluster

Files. `ns_compass.py` (everything) and `run_ns_compass.slurm` (example job). All results go into one flat folder `ns_results/`:

- `ns_N<N>_lam<..>_k2<..>.npz`: all numbers for one N;
- `summary_lam<..>_k2<..>.txt`: the printed report, appended per N;
- `summary_table_lam<..>_k2<..>.txt`: one line per N;
- `fig1_...png` to `fig5_...png`: the figures shown in this note.

Usage. From the folder that contains `ns_compass.py`:

```
python ns_compass.py --N 2 3 4 5 6 7 8 --lam 0.6 --kappa2 0.05 \
                     --workers 8 --mem-budget 110          # thesis
python ns_compass.py --N 2 3 4 5 6 7 8 9 10 11 12 13 14 --lam 1.0 --kappa2 0.5 \
                     --workers 8 --mem-budget 110          # scaling
python ns_compass.py --plot-only --lam 0.6 --kappa2 0.05   # figures only
sbatch run_ns_compass.slurm                                # thesis, cluster
sbatch --export=ALL,PRESET=scaling run_ns_compass.slurm    # scaling, cluster
```

Other options: `--wc`, `--wa`, `--M` (explicit cutoffs, one per N). Default parameters are at the top of the script. Results of separate jobs with the same λ, κ_2 are combined by `--plot-only`, so large N can run in their own jobs.

Cost. The table gives the script's own memory estimate per block, how many of the six blocks it then runs at once with `--mem-budget 110`, and a rough wall time scaled from the test machine (factor ~ 2 uncertainty). In the test runs the measured peak per block was about half the estimate at $\lambda = 0.6$–0.7, $\kappa_2 = 0.05$, and within 15 % of it at $\lambda = 1$, $\kappa_2 = 0.5$ ($N = 6$–8). Smaller N take minutes.

preset	N	M	k^*	block dim.	GB/block	parallel	wall time
thesis ($\lambda = 0.6$, $\kappa_2 = 0.05$)	6	176	1	1.9×10^5	17	6	10 min
	7	196	1	3.1×10^5	37	2	1 h
	8	216	1	4.7×10^5	75	1	4.5 h
scaling ($\lambda = 1$, $\kappa_2 = 0.5$)	11	106	3	2.0×10^5	19	5	15 min
	12	114	4	2.7×10^5	31	3	30 min
	13	120	4	3.5×10^5	46	2	1.3 h
	14	126	4	4.5×10^5	68	1	4 h

$N = 9$ at the thesis parameters would need ~ 140 GB per block (estimate) and does not fit. SuperLU stores indices as 32-bit integers, which also caps the LU at about 2×10^9 non-zeros per factor.

Reading the report. For each N the script prints: the block-structure check (must be 0); per block the LU size, time and peak memory; the Fock-tail check; the vacuum cut; the selected mode per block with its eigenvalue and the next SR mode (gap); the qubit-separation diagnostics; the NS test for qubits A and B; the relation diagnostics; the which-qubit memory versus spin flips with k^*; the cut scan.

If something goes wrong.

- *"a worker process died"*: out of memory; lower `--workers` or `--mem-budget`, or request more memory.
- *"ARPACK NOT CONVERGED"*: rare; increase `N_MODES` or move `SIGMA`.
- *"INCREASE M"*: Fock tail too large; raise `M_SIGMAS` or pass `--M`.
- `n_cut = 0`: no vacuum valley was found (vacuum negligible, or not separated from the lobe, typical for small photon numbers); look at `fig4` of the run (as Fig. 4 here).

10 Limitations

- Exact diagonalisation limits N to $\lesssim 8$ (thesis parameters) or $\lesssim 14$ ($\lambda = 1$, $\kappa_2 = 0.5$) on a 128 GB node. The thermodynamic-limit statement rests on the trend with N and on the mechanism of Sec. 7, not on a single large-N point.
- Eigenvectors are accurate to $\sim 10^{-10}$; NS distances below that are not meaningful.
- The vacuum cut is a sharp projector. Its effect is monitored by the cut scan but not removed; it matters most at the smallest N and photon numbers.
- "Steady" means slowest mode in the block. At finite N these modes decay slowly; the script reports their rates so the degree of metastability is always visible.